

B. Sc. 5th Semester (Honours) Examination, 2025

Computer Science

Course ID: 51512

Course Code: SH/CSC/502/C-12

**Computing Theory (Theory)
(NEW CBCS Syllabus 2022-23)**

Time: 2 Hours

Full Marks: 40

The figures in the margin indicate full marks

UNIT I

1. Answer any 5 (five) of the following questions:

(5 × 2 = 10)

- a) What do you mean by transition diagram of FA?
- b) Differentiate sentences vs. sentential forms?
- c) Define FA with ϵ transition.
- d) Find the set denoted by regular expression $(a|b)(a|b)$.
- e) State pumping lemma for regular languages.
- f) Define Turing Machine.
- g) Define CNF.
- h) Define ambiguous grammar.

[Turn Over]

UNIT II

2. Answer any four of the following questions:

(5 × 4 = 20)

a) Prove by method of induction that $n^3 + (n+1)^3 + (n+2)^3$ is always divisible by 3. What do you mean by deductive proof? 4+1=5

b) Draw nfa for $L = (a+b)^*abb$. convert this nfa into equivalent dfa. What is proof by contradiction? 1+3+1=5

c) Design a pda for $L = \{ ww^R : w \in (a+b)^* \}$. What is instantaneous description? 4+1=5

d) Design a Turing machine for $L = \{ a^{2n}b^n : n > 0 \}$. 5

e) Write context free grammars for $L_1 = \{ ww^R : w \in (a+b)^* \}$ & $L_2 = \{ a^n b^m : n \neq m \}$ 2.5×2=5

f) Define ϵ -closure of a state in finite automata? Give example. Identify the language generated by following grammar, where S is the Start variable.

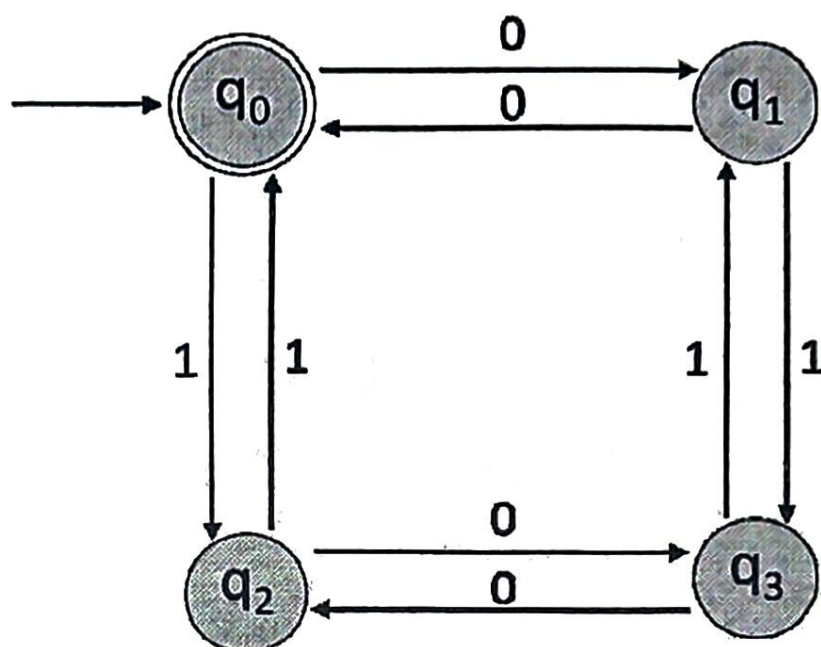
$\{ S \rightarrow XY, X \rightarrow aX \mid a, Y \rightarrow aYb \mid \epsilon \}$ State any two properties of CFL's 2+2+1=5

UNIT III

3. Answer any 1 (one) of the following questions:

(10 × 1 = 10)

a) Find a regular expression for the following dfa.



Show that $L = \{a^p / p \text{ is prime}\}$ is not a regular language.

5+5=10

b) Convert the following CFG into equivalent GNF:

$S \rightarrow AB$, $A \rightarrow BS|1$, and $B \rightarrow SA|0$. Prove that regular

languages are closed under Union & intersection. What do you mean by transducer? Give example.

4+4+2=10

B. Sc. 5th Semester (Honours) Examination, 2025
Computer Science

Course ID: 51512

Course Code: SH/CSC/502/C-11²

Theory of Computation (Theory)
(OLD CBCS Syllabus 2017-18)

Time: 2 Hours

Full Marks: 40

The figures in the margin indicate full marks

UNIT I

1. Answer any 5 (five) of the following questions:

(5 × 2 = 10)

- a) What do you mean by transition diagram of FA?
- b) Give example of two regular languages.
- c) Define FA with & transition.
- d) Find the set denoted by regular expression $(a|b)(a|b)$.
- e) State Arden's theorem.
- f) State pumping lemma for regular languages.
- g) Give example of two context free languages.
- h) Define PDA.

[Turn Over]

UNIT II

2. Answer any four of the following questions:

(5 × 4 = 20)

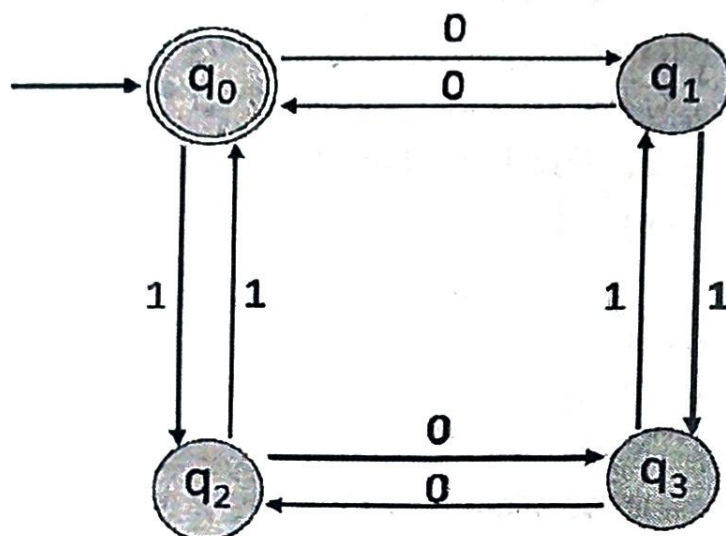
- a) Prove by method of induction that
 $1^2 + 2^2 + 3^2 + \dots n^2 = [n(n+1)(2n+1)]/6$ 5
- b) Draw nfa for $L = (a+b)^* abb$. convert this nfa into equivalent dfa. What is proof by method of contradiction? 1+3+1=5
- c) Design a pda for $L = \{a^n b^n : n > 0\}$. What is instantaneous description? 4+1=5
- d) Design a Turing machine for $L = \{a^n b^n : n > 0\}$.
- e) Using pumping lemma prove that $L = \{a^n b^n : n \geq 0\}$ is not regular. 5
- f) Define ϵ - closure of a state in finite automata? Give example. Identify the language generated by following grammar, where S is the Start variable.
 $\{S \rightarrow XY, X \rightarrow aX \mid a, Y \rightarrow aYb \mid \epsilon\}$ State any two properties of CFL's 2+2+1=5

UNIT III

3. Answer any 1 (one) of the following questions:

(10 × 1 = 10)

a) Find a regular expression for the following dfa.



Draw NFA for following languages-

i. $a(a+b)^*b$.

ii. a^*b^*

5+5=10

b) Convert the following CFG into equivalent GNF:
 $S \rightarrow AB$, $A \rightarrow BS|1$, and $B \rightarrow SA|0$. Prove that regular

languages are closed under Union & complementation.

What do you mean by acceptor? Give example. $4+4+2=10$

Bankura Sammilani College
Department Of Computer Science
Internal Assessment Examination - 2024
Semester – V (Theory of Computation)

Time 30 Minutes

F.M – 10

Answer any 5 questions-

5x2=10

- ☒ Define ϵ – closure of a state in finite Automata.
- 2. What is LL grammar? Give example.
- ☒ Give a regular expression for all strings of $\{a, b\}$ which starts with ab and ends with bb .
- ☒ State Arden's theorem.
- ☒ Draw an NFA for a^*b^* .
- ☒ State Pigeonhole principle.
- ☒ Write pumping lemma of regular languages.

aaa bbb
ababab

